

TrES-2b

R-1571

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_140808 · created 2026-08-01T22:43:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456877.90089 +/- 0.00031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.048 +/- 0.043
Transit depth (Rp/Rs)^2	0.23 +/- 0.41 [%]
Orbital Inclination (inc)	80.59 +/- 2.86
Airmass coefficient 1 (a1)	1.12 +/- 0.16
Airmass coefficient 2 (a2)	-0.08 +/- 0.061
Scatter in the residuals of the lightcurve fit is	13.44 %
Best Comparison Star	#3 - [26, 136]
Optimal Aperture	1.94
Optimal Annulus	10.82
Transit Duration (day)	0.016 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.048 $\pm$ 0.043 vs 0.1254 (deviation 1.8 σ)
Duration fit vs published	0.016 d vs 0.0746 d (deviation 2.02 σ)
Mid-time O-C	0.5 min
Depth SNR	1.1
Residual scatter	13.44 %

Light curve

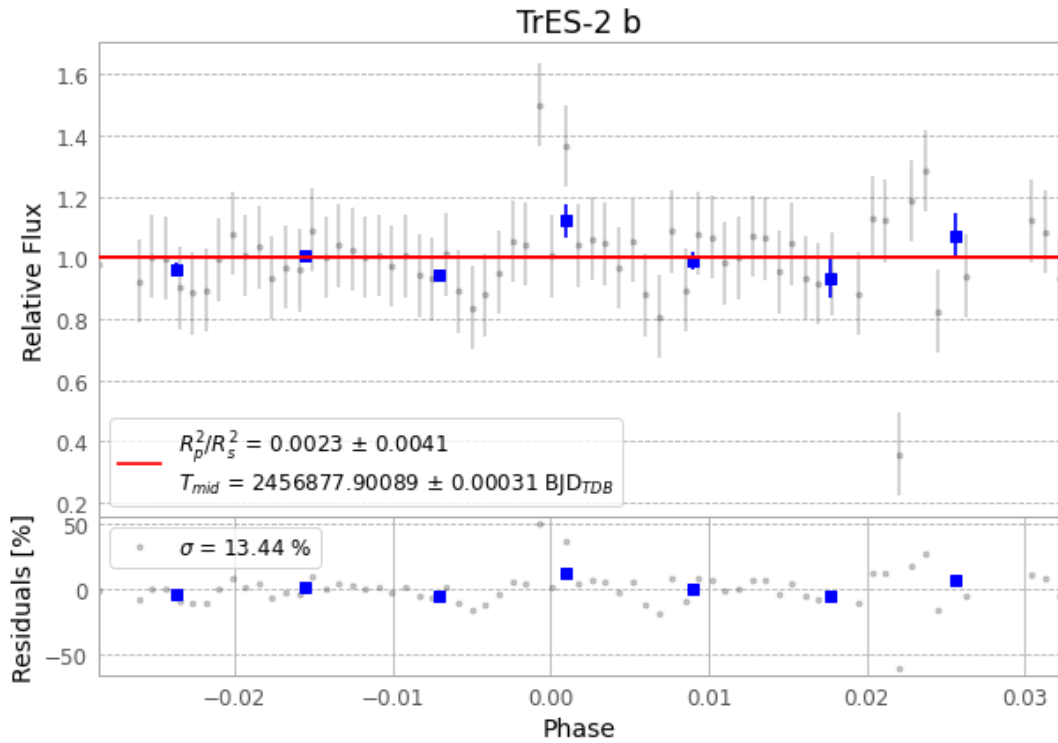


Figure 1 — FinalLightCurve_TrES-2 b_2014-08-08.png (SHA-256 0c244c777b034974...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [291, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 24aaabd5dd4d9ef7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), TRES-2140808075113.FITS → TRES-2140808113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-140808.log (69d775fcb247...), FinalLightCurve_TrES-2 b_2014-08-08.png (0c244c777b03...), FinalLightCurve_TrES-2 b_2014-08-08.csv (e305063e78a4...)

Compute resources

Wall time 350.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 22:43Z.